

Personal Statement

My understanding of technology began with the experience of assembling a computer by myself. When the scattered hardware components came together through precise collaboration to form an integrated whole, I not only understood the interaction logic at the physical level but also developed a strong interest in the intelligent systems that drive the operation of the hardware. This understanding continued to be deepened throughout my subsequent learning from website development to server operation and maintenance. This also enabled me to gradually acquire the ability to build complex systems, and made me realize that artificial intelligence is reshaping the paradigm of business innovation. Moreover, when I see technologies like AI tools and smart cars turning human imagination into reality, I am even more eager to stand at the forefront of this transformation. My goal is to turn technology, from an entrepreneur's perspective, into products and services that truly improve people's lives, more than just participating in the design of intelligent systems as a developer; it also includes.. The combination of AI and business represents the future direction of technological innovation, and it also carries my original aspiration to create the future.

During my undergraduate studies, I systematically built a knowledge system in information technology and AI, and deepened my understanding through the integration of theory and practice. Courses such as Introduction to Software Engineering and Software System Design and Architecture have enabled me to master the overall process of the software development lifecycle, as well as understand the logic of system architecture design, modular management, and network communication. These courses taught me how to analyze problems from a system perspective, design efficient and scalable solutions, and maintain structured thinking in complex projects. In courses closely related to AI, such as *Data Structures and Algorithms*, and *Artificial Intelligence Foundation*, I gradually developed an algorithm-centered way of thinking and understood the entire process from data input and model training to result output. These courses strengthened my abilities in logical reasoning, computational complexity analysis, and algorithm optimization. They also allowed me to understand the working principles of AI systems from the underlying mechanisms.

Beyond the classroom, I continued to deepen my knowledge through project practice. In a traffic inquiry system developed with classmates, I was responsible for frontend development based on the Vue framework. Faced with a completely unfamiliar framework, I read official documents, analyzed open-source projects, and used AI tools for learning. I quickly mastered its component-based logic and data interaction mechanisms, and successfully completed the development task. This experience made me deeply appreciate the mutual promotion between theory and practice: theory provided me with ideas to solve problems, while practice allowed me to verify and optimize these ideas in real scenarios. To further consolidate my knowledge, I built my own GitHub repository, which systematically records the source code of projects I participated in, study notes, and development insights. This facilitated the creation of a sustainable learning system while enabling me to rapidly learn and innovate with new technologies. What's more, during a one-month server operation and maintenance internship, I gained a deep understanding of the integrity and complexity of computer systems. My main task involved developing an auto-update script for a bastion host (JumpServer) to facilitate its automatic "drift" to a new server upon failure. Initially, I planned to detect the new environment by checking for mismatched system IDs in configuration files. However, I discovered these files updated automatically during migration, rendering this method ineffective. After multiple attempts and debugging, my solution was to have the script, triggered at system startup, query the cloud provider's API using the machine's ID to obtain its purchase time. If this time was very recent (e.g., within minutes) compared to the script's execution time, it confirmed the environment shift, prompting the script to execute the necessary configuration updates. This process not only successfully ensured the migration but also made me realize that the various modules of a system are deeply interconnected, not isolated. At the same time, when writing backup scripts for the bastion host, I accidentally deleted configuration files due to an operational error. This mistake made me acutely aware of the critical importance of permission management and risk control in system design and operation. More importantly, whenever I encountered unsolvable errors and sought help from seniors, they would not give me the answer directly. Instead, they

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guided me to track the root cause of the problem by recording logs and analyzing execution paths. This problem-oriented solution helped me gradually master methods for independent analysis, iterative improvement, and experience summarization.

This internship experience not only deepened my understanding of system operation and maintenance but also shifted my perspective from engineering to commercial and management thinking. I realized that technological innovation can only be transformed into sustainable value creation when combined with management mechanisms, risk control, and resource integration. During the process of writing scripts and optimizing workflows, I constantly thought about how to improve efficiency and reduce error rates—this is essentially a form of micro-innovation in production processes and a reflection of entrepreneurial thinking at a micro level. I tried to solve systematic problems with limited resources and create higher organizational efficiency. From this, I gradually understood that management is not just about operating a company but also about transforming technological potential into practical productivity. This experience equipped me with the ability to identify problems from practice, integrate resources, and drive improvements. It also laid a solid practical foundation for my study of business-related content.

Triggered by this, I have clarified my goal, integrating technological potential into the improvement of practical productivity, considering the MSc Artificial Intelligence in Business program at the City University of Hong Kong as an ideal platform for further education. I think that the program's unique curriculum setting precisely targets the ability dimension that I need to strengthen the most. I intend to systematically apply cutting-edge technologies, such as artificial intelligence and machine learning, to real business environments. I intend to particularly focus on the expertise in the courses Artificial Intelligence for Business Applications and AI-Enhanced Business Analytics with Excel and Python. I believe that the former will help me establish a complete knowledge framework for the intelligent decision-making system, especially in key business scenarios such as demand forecasting and resource optimization. The latter will enable me to master the entire process from data collection to business insights, which is precisely the capability dimension that I need to strengthen in my previous experiences. In summary, through systematic learning of methodologies such as business scenario modeling and data-driven decision-making, I will be competent in effectively applying machine learning technology to solve practical business problems. The combination of this technical ability with business thinking will significantly empower me to undertake the work of developing intelligent business systems, truly achieving the leap from technical execution to business innovation.

Under the combined effects of my systematic academic training and diverse practical experiences, I have a clear plan for my future career direction. After graduation, I plan to join an AI technology-driven enterprise, working on AI system development and innovative product design to accumulate comprehensive experience in technology R&D and project management. Within 3 to 5 years, I hope to establish an innovative startup company centered on AI technology, focusing on applying AI to the intelligent solution of real-world problems, such as urban traffic optimization, enterprise information security, and automated system construction.



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Recommendation Letter

Date: September 23, 2025

To whom It May Concern,

It is my pleasure to write this recommendation letter for LUO Xiaoyu, one of my best students in the course *Introduction to Computer Science and Technology*, to support him in applying for this esteemed postgraduate program at your university.

I came into acquaintance with Xiaoyu in September 2022 in my class, during which he fully showed his intelligence and concentration in class, behaving in a friendly and constructive manner in inner-group cooperation. In the process of interacting with him, both inside and outside the classroom, I have come into acquaintance with his solid theoretical foundation and problem-solving ability. He demonstrated his systematic understanding of the parts like computer system architecture, data representation, algorithm principles, programming languages, equipping himself with a fundamental cognitive framework of computer science. He also actively participated in diverse algorithm design and programming practices involving network programming, concurrent programming, and performance optimization, to cultivate his ability tailored to solve complex problems.

As I recalled, he took the initiative to participate in numerous programming practices, during which he wrote simple programs to analyze the machine-level representation of data and the process of instruction execution. He was able to use the GDB debugging tool to track the changes of program variables and understand the correspondence between high-level languages and machine instructions. Simultaneously, he was also adept at precisely reverse-analyzing the binary programs, and deeply studied the buffer overflow attacks and program linking mechanisms. Such things enhanced his understanding of the underlying principles of computer systems and refined his competencies in programming and system design, earning him a good score--86.

In a nutshell, it is without any reservation that I recommend this outstanding individual for admission to your esteemed program. I believe that he will continue to make positive contributions to this program, and the academic environment and lived experience in your university will be very valuable to his development.

If you need further information, please feel free to contact me.

Sincerely

Signature: